RABIUL ISLAM

AI / MACHINE LEARNING ENGINEER | GENERATIVE AI · NLP · COMPUTER VISION

Bangladesh | GitHub: github.com/rabiul9137 | LinkedIn: linkedin.com/in/rabiulislam01

SUMMARY

AI/ML and Data Science professional with an MSc in CSE (Data Science) and 10+ years in technology and network infrastructure. Builds Python-based generative AI (RAG, tool-calling and multi-agent systems), NLP, and computer vision applications, and packages them as APIs with FastAPI and Docker. Holds PMP and Scrum Master certifications and uses Git/GitHub for version control. Seeking an AI/ML Engineer role delivering practical, well-engineered AI solutions for clients.

TECHNICAL SKILLS

Generative AI & NLP: LLM applications, RAG, AI agents and multi-agent workflows, prompt engineering, embeddings, vector search, Hugging Face Transformers, BERT, DistilBERT, TF-IDF, Word2Vec, PEFT/LoRA
ML & Deep Learning: supervised/unsupervised learning, classification, regression, clustering, feature engineering, model evaluation, neural networks, CNNs, transfer learning, PyTorch, TensorFlow
Programming & Data: Python, SQL, R, VBA, NumPy, pandas, scikit-learn, Matplotlib, ETL, Power BI, Tableau
Computer Vision: image classification, CNN-based vision, vision pipelines, multimodal and vision-language model concepts
Engineering & Delivery: FastAPI, REST APIs, Docker, Git/GitHub, Agile/Scrum, Jupyter, Google Colab, Kaggle, VS Code
Networking & Infrastructure: LAN/MAN/WAN, Cisco switches, MikroTik routers, Windows/Linux, data-center and server operations, network security

PROJECTS

Bangla Knowledge-Base Chatbot (RAG)

- Built a Bengali knowledge-base chatbot that retrieves source passages and supports chapter/section-based citations.

Leo: Multi-Agent AI Tutor | GitHub: github.com/rabiul9137/leo-multi-agent-ai-tutor

- Built a multi-agent AI tutor with CrewAI, Groq, Pydantic, and Streamlit, providing personalized lessons, automated quizzes, answer evaluation, and student memory.

Medical Dataset Multi-Tool AI Agent

- Developed an agent workflow to query medical datasets and use web search/tools; worked with heart, cancer, and diabetes datasets in a course project.

Bangladeshi Banknote Detection API | GitHub: github.com/rabiul9137/taka-note-detection-api

- Built a YOLO-based computer-vision model/API to detect and classify Bangladeshi banknotes; exposed prediction through FastAPI and packaged the service with Docker.
- Tested inference on sample note images; recorded a Ten Taka detection at 0.753 confidence. (A single test result, not a general model-performance metric.)

IMDB Movie Review Sentiment Analysis | GitHub: github.com/rabiul9137/sentiment-analysis-imdb

- Compared TF-IDF + Logistic Regression, Word2Vec + Logistic Regression, and DistilBERT on 50,000 reviews; achieved 90.78% accuracy and 91.01% F1 with DistilBERT.

TriageMindai: Explainable, Bias-Aware ED Triage | GitHub: github.com/rabiul9137/TriageMindai

- Built an explainable, bias-aware AI system for emergency-department triage using clinical data, NLP, and machine learning.

UIU Freshers AI Guide (in development) | GitHub: github.com/rabiul9137/uiu-freshers-ai-guide

- Developing a FastAPI-backed information assistant using curated UIU information; current knowledge base is being expanded and tested.

EXPERIENCE

Assistant System Engineer — United International University | Feb 2014–Present

- Maintain university data-center, server, network, and IT infrastructure, supporting system availability and reliable operations.
- Administer LAN/MAN/WAN, Cisco switches, and MikroTik routers; troubleshoot Windows/Linux systems and infrastructure issues.
- Monitor infrastructure performance, document technical solutions, and coordinate IT support activities.

EDUCATION & CERTIFICATIONS

MSc, Computer Science & Engineering (Data Science) — United International University |
BSc, Computer Science & Engineering — Bangladesh University of Business & Technology |
Diploma in Engineering, Computer Technology
Certifications: CCNA; CCNA Security; PMP; Scrum Master